

Day 4 Report — P6-C

Date: 22/09/2026

General Progress

Today was mainly focused on finishing the main parts of the project and making sure the different components were ready to work together. The database, backend, UI, mobile application, testing, and AI parts all reached their final stages.

The **PostgreSQL database was fully set up with both the student and workshop seed data**, and the database structure, columns, and IDs were reviewed. The backend server was also successfully connected to the database and is now running with the required APIs. The Flutter handoff files were prepared with the API documentation, authentication details, tokens, and endpoint permissions.

On the AI side, the student performance dataset was finalized and the machine learning pipeline was completed. The UI and mobile application were also completed, while testing covered the main system flows, security, and different user roles.

Individual Feedback

Arwa — Squad Lead & Data Analyst

I continued following up with the team and worked on the AI/data part of the project. I finalized the **student performance dataset and completed the machine learning pipeline**, including the data preprocessing, feature preparation, model training, and evaluation. I also prepared the final trained model so it can be used later as part of the system.

Helana — UI/UX / Frontend

Helana completed the **UI work** and continued working with Samuel to make sure the final mobile application follows the same design. They also finalized the color scheme and made sure the UI and actual app implementation stay consistent.

Samuel — Mobile App Developer

Samuel completed the **mobile application**, including the different screens, navigation, and button interactions. The app is now ready to be connected with the backend APIs, with the necessary changes already prepared on the application side for the backend connection.

Felopateer — Software Testing / QA

Felopateer completed and documented the **Test Cases and security rules** for the system. His work covered authentication, RBAC, unauthorized access, privilege escalation, Separation of Duties, data scopes, Job Cards, inventory and stock movement, student assessments, and certificates.

He also **helped throughout the project, especially with the backend**, and supported Abdelmasih with checking and testing different parts of the backend and system functionality.

Abdelmasih — Backend Developer

Abdelmasih completed a major part of the backend and database setup. He prepared the **PostgreSQL database**, including the schema, student seed data, and workshop seed data, and reviewed the columns and IDs to make sure everything was consistent.

The backend server is now running on the local network and the **health check is working**. He also increased the number of APIs and prepared the `flutter_handoff` folder containing the `openapi.yaml`, authentication and token explanation, and the endpoints with their required permissions.

He also created **12 test accounts**, one for each role, with different passwords, and added them to the schema so they are created automatically with the database. The login was tested using `manager@test.com` and returned a valid access token, confirming that the authentication is working end-to-end.

He also made the required changes in the Flutter project, including the API base URL, sending the email instead of the username, handling errors correctly, and adding the required internet/HTTP permissions for Android and iOS.

Team Members Contribution

Everyone contributed throughout the project and the work became much more connected during the final stages. The database and backend were developed alongside the UI, mobile application, testing, and AI work, and Felopateer also supported the backend work in addition to his testing responsibilities.

By the end of the project, we had the **database and seed data ready, the backend running with APIs and authentication, the UI and mobile app completed, the AI model prepared, and the main testing and security cases documented**.

Overall Feedback

The team made good progress throughout the project and managed to bring the different parts together during the final stage. One of the most important achievements was getting the backend, database, authentication, and mobile application prepared for integration instead of having each part working separately.

There was also good communication between the team members, especially between **backend, mobile development, testing, and UI/UX**, which helped with fixing issues and keeping the different parts aligned.

The main next step is to complete the **final integration testing between the Flutter application and the backend**, and make sure the main workflows work correctly from the user's side.

Day 4 status: Main components are completed → moving into final integration and testing.